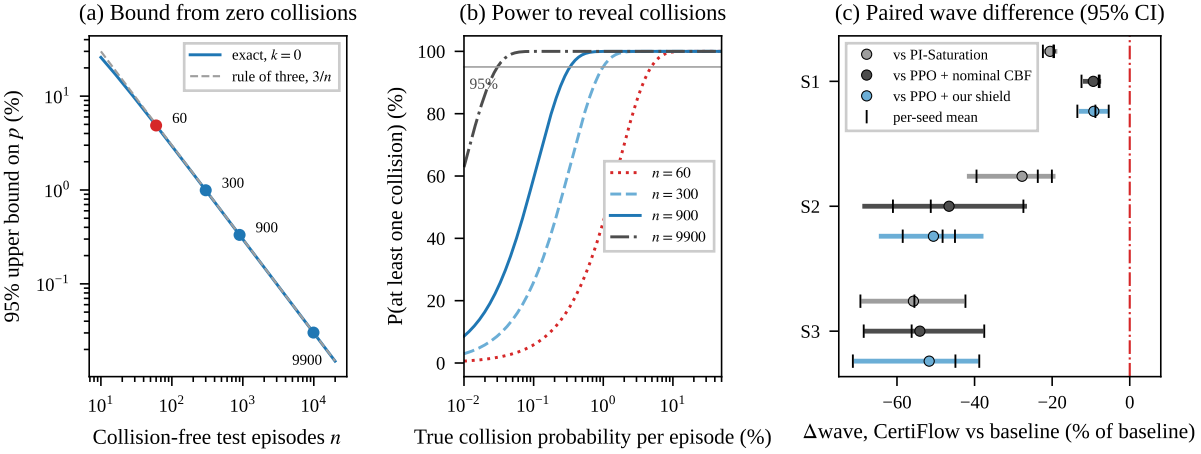
(a) Bound from zero collisions
exact, $k = 0$
rule of three, $3/n$
60
300
900
9900
95% upper bound on p (%)
Collision-free test episodes n
(b) Power to reveal collisions
95%
$n = 60$
$n = 300$
$n = 900$
$n = 9900$
P(at least one collision) (%)
True collision probability per episode (%)
(c) Paired wave difference (95% CI)
vs PI-Saturation
vs PPO + nominal CBF
vs PPO + our shield
per-seed mean
S1
S2
S3
Δwave, CertiFlow vs baseline (% of baseline)